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Petro Refinery LLC: Linear Programming Exercise— Tutorial Handout

Petro Refinery LLC, a crude oil refinery, faces an issue common in many production and service industries: finding the optimal product mix or input mix. A central difficulty is that each product consumes different amounts of the firm's shared resources, but contributes different revenues toward the firm's fixed costs. Managers do not necessarily want to sacrifice the production of one product for another just because one has a higher contribution.ⁱ Even so, the best utilization of the firm's resources toward a good product mix should drive the firm's marketing strategy—the mix of products that will result in the most profit. Petro Refinery is a simple but realistic exercise to illustrate product mix decisions.

Setting Up Petro's Problem

Petro Refinery must decide the optimal number of barrels of light and heavy crude oil to process each week to *maximize contribution toward fixed costs*. Heavy and light crude form our problem's *decision variables* — the variables within a model that the firm can control. In our case, the variables represent the number of barrels of light and heavy crude to be processed:

- X_1 = Number of barrels of light crude to process (in thousands)
- X_2 = Number of barrels of heavy crude to process (in thousands)

The contribution margin for processing one barrel of light crude is \$10; it is \$12.5 for heavy crude. Each product's contribution margin is the decision variable's *objective coefficient*. The *objective* is to maximize total contribution. The total contribution from processing (in U.S. thousands) is therefore given by:

$$10X_1 + 12.5X_2$$

The refinery process uses three main pieces of equipment: distillation, conversion, and blending. These resources have limited availability, and therefore impose *constraints* on the amount of crude that can be

ⁱ Contribution is the amount of money available to put towards the firm's fixed costs. Once fixed costs are covered, contribution is the amount available as profit (i.e., contribution - fixed costs - profit).

Published by WDI Publishing, a division of the William Davidson Institute (WDI) at the University of Michigan.

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processed. Creating 1,000 barrels of either light or heavy crude will require 2 hours of distillation for each type. Therefore, the total number of distillation equipment hours required is given by: $2X_1 + 2X_2$. The distillation equipment is available only for 3,000 hours each week, and we can indicate that as:

$$2X_1 + 2X_2 \leq 3,000$$

Similar to distillation equipment, we can express the conversion constraint as:

$$2X_1 + 3X_2 \leq 2,400$$

and the blending constraint as:

$$3X_1 + 2.5X_2 \leq 2,400$$

We also require a non-negativity constraint, so that our linear programming problem (LP) will not allow negative values to exist in our solution. The non-negativity constraint looks like: $X_1 \geq 0$ and $X_2 \geq 0$. Therefore, Petro's LP of solving for the optimal barrel production that maximizes contribution is set up as:

$$\text{Maximize: } 10X_1 + 12.5X_2$$

Subject to:

$$\text{Distillation constraint: } 2X_1 + 2X_2 \leq 3,000$$

$$\text{Conversion constraint: } 2X_1 + 3X_2 \leq 2,400$$

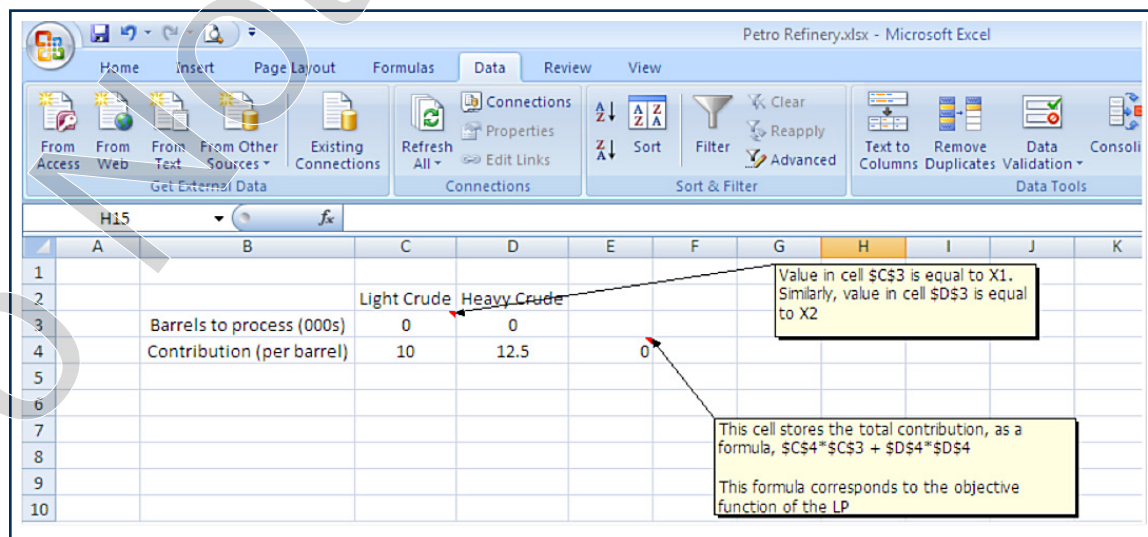
$$\text{Blending constraint: } 3X_1 + 2.5X_2 \leq 2,400$$

$$\text{Non-negativity constraints: } X_1 \geq 0, X_2 \geq 0$$

Setting Up an LP Model in Excel

The next step is to use Microsoft Excel to solve the problem. After opening Excel, enter the *decision variables* and the *objective coefficients* into cells as in **Figure 1**, where the decision variables X_1 and X_2 are stored in \$C\$3 and \$D\$3, and the objective coefficients in \$D\$3 and \$D\$4.

Figure 1
Decision Variables and Objective Function



We can then express the LP's *objective function* (total contribution, in this case) for any given decision variable value in Excel terms as $\$C\$4 * \$C\$3 + \$D\$4 * \$D\3 . The objective function is stored in cell $\$E\4 as a *formula*, which allows the solver to evaluate the objective function across a wide range of variable values.

Constraints

The distillation constraint will be stored in $\$C\10 and $\$D\10 (see **Figure 2**). For a given decision variable (stored in $\$C\3 and $\$D\3), the total number of distillation equipment hours is given by $\$C\$10 * \$C\$3 + \$D\$10 * \$D\3 . The distillation equipment usage is stored as a formula in $\$E\10 .

The distillation equipment hours used (cell $\$E\10) cannot exceed the available hours (3,000, cell $\$G\10). (The $\leq$ in $\$F\10 notates that the value calculated by the formula in $\$E\10 should be no more than the value of $\$G\10). The other constraints, conversion and blending, can be expressed in a similar manner as shown in **Figure 3**.

Figure 2
Setting Up the Distillation Constraint

	A	B	C	D	E	F	G	H	I	J	K
1											
2			Light Crude	Heavy Crude							
3		Barrels to process (000s)	100	0							
4		Contribution (per barrel)	10	12.5	1000						
5											
6											
7											
8		Constraints									
9					Used		Available				
10		Distillation	2	2	200	<=	3000				
11											
12											
13											
14											
15											
16											

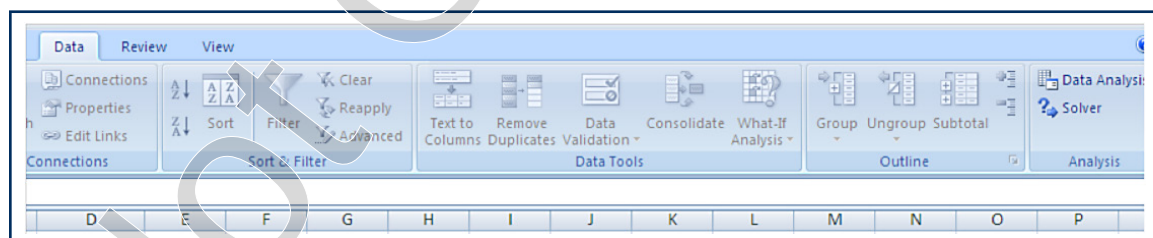
Figure 3
Setting Up All Constraints

	A	B	C	D	E	F	G	H
1								
2			Light Crude	Heavy Crude				
3		Barrels to process (000s)	100	0				
4		Contribution (per barrel)	10	12.5	1000			
5								
6								
7								
8		Constraints						
9					Used		Available	
10		Distillation	2	2	200	<=	3000	
11		Conversion	2	3	200	<=	2400	
12		Blending	3	2.5	300	<=	2400	
13								

Using the Solver

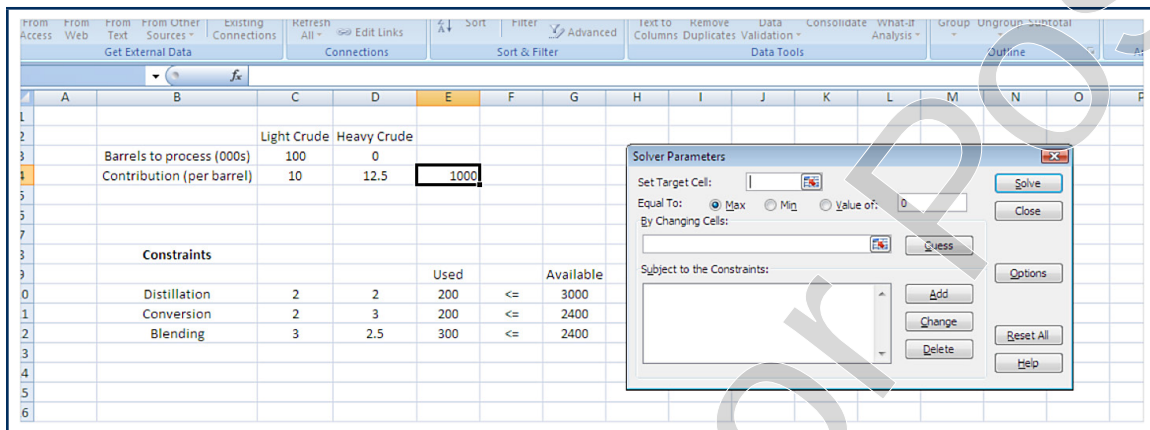
After setting up the problem, click on the “Solver” tool under the “Data” tab (see **Figure 4**). (This might involve a slightly different process if the computer is a Mac instead of a PC.) If you do not see the Solver option under the Data tab, you need to install the Solver Add-in. You can do this by clicking on the Microsoft Office Button, then “Options,” then “Add-Ins,” then “Go,” and then selecting “Solver” from the menu of options.

Figure 4
Solver Tool in Excel



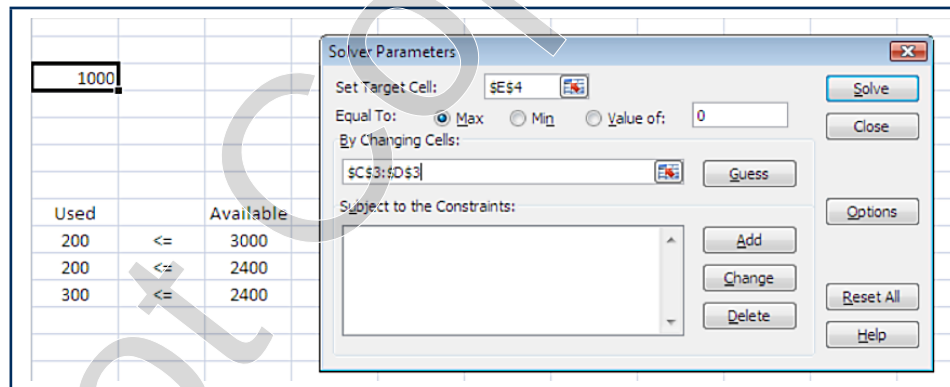
Clicking on the Solver option will open a dialog box as shown in **Figure 5**.

Figure 5
Solver Parameters Dialog Box



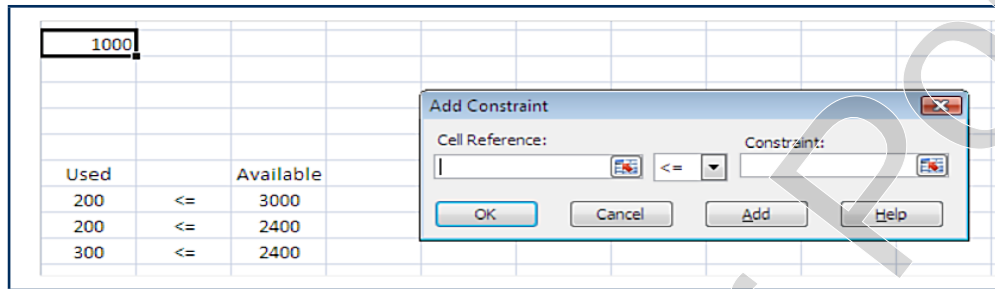
The target cell (\$E\$4) corresponds to the cell storing the objective function value. Because we are interested in maximizing total contribution, we select the “Max” button. The changing cells correspond to the decision variable cells (\$C\$3:\$D\$3). Solver will vary the values in these cells in order to achieve the Max value in the target cell. The Solver Parameters dialog box should look like **Figure 6** at this stage.

Figure 6
Solver Input Window with Target Cell and Changing Cells Populated



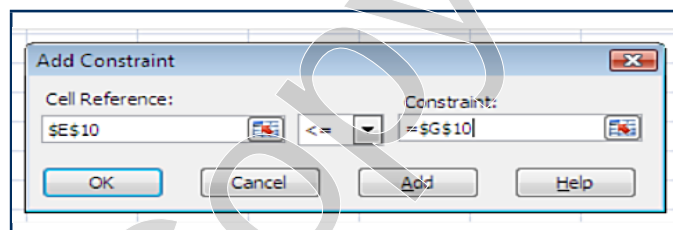
To specify the constraints we click on the “Add” button next to the constraints box, which will open a dialog box as in **Figure 7**.

Figure 7
Adding Constraints to Solver



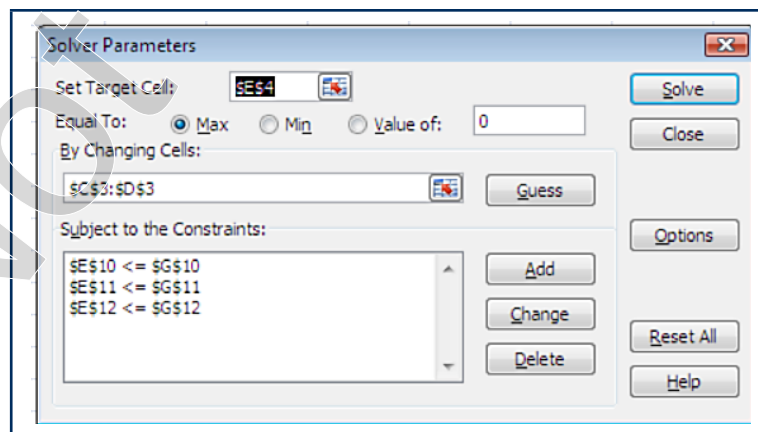
“Cell Reference” corresponds to the amount used of the resource, while “Constraint” corresponds to the available amount of the resource. For distillation, the Cell Reference should be \$E\$10 and the Constraint should be \$G\$10. Clicking on “Add” will add this constraint to the set of constraints. The conversion and blending constraints can be added in a similar manner (see **Figure 8**).

Figure 8
Distillation Equipment Constraint



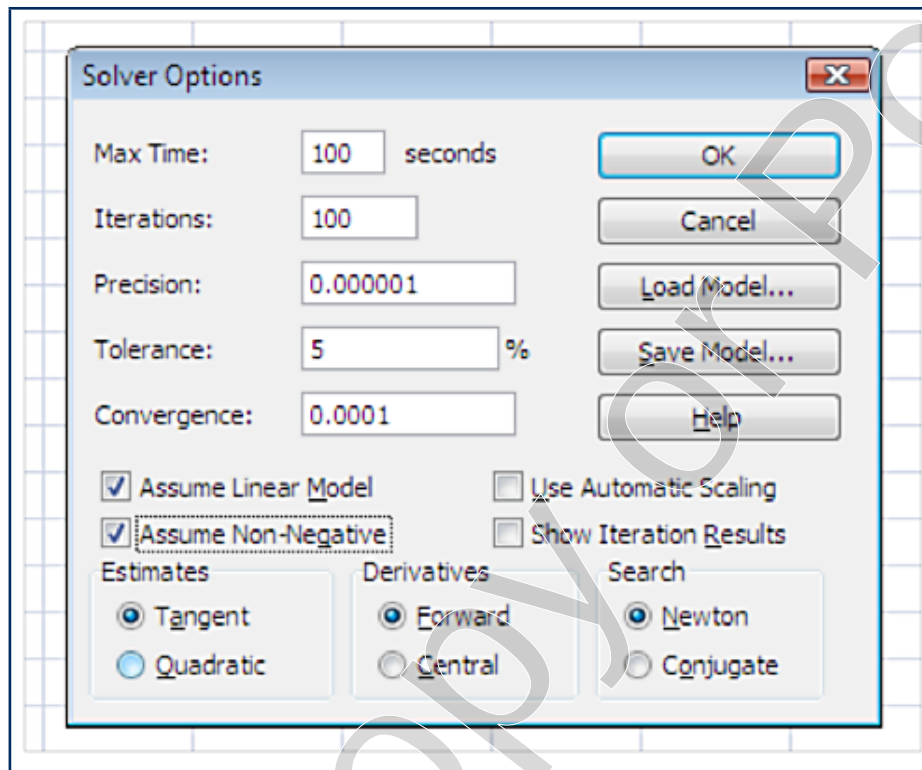
The Solver Parameters dialog box should look like **Figure 9** once all constraints have been added.

Figure 9
All Resource Constraints



As a final step, we need to specify that this is a linear program. We do this by clicking on “Options” and selecting “Assume Linear Model” in the dialog box. We also select “Assume Non-Negativity” to impose the non-negativity constraints for the LP (see **Figure 10**).

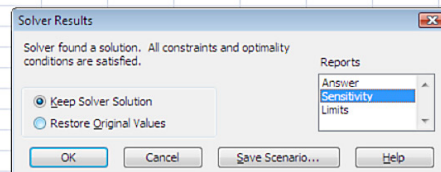
Figure 10
Solver Options to Specify Linear Program and Non-Negativity Constraints



The model is now fully specified. Click on “Solve” to solve the problem. **Figure 11** shows the output.

Figure 11
Solver Output

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
		Light Crude	Heavy Crude											
	Barrels to process (000s)	300	600											
	Contribution (per barrel)	10	12.5	10500										
	Constraints			Used		Available								
	Distillation	2	2	1800	<=	3000								
	Conversion	2	3	2400	<=	2400								
	Blending	3	2.5	2400	<=	2400								



Cells \$C\$3 and \$D\$3 are now populated with the optimal values; the maximum contribution is achieved by processing 300,000 barrels of light crude and 600,000 barrels of heavy crude each week. Further, \$E\$4 shows the corresponding contribution (\$10,500,000 per week), and \$E\$10:\$E\$12 shows the resource usage. At these production rates, the conversion and blending resources are completely utilized (since “used” values equal the “available” values), but only 60 percent, of the distillation is used (1,800 is used out of an available 3,000).

Solver also provides additional information that can be useful for making managerial decisions. For instance, what would be the value to Petro if it could add conversion equipment hours? Questions such as these can be answered using a Solver-generated *sensitivity report*.

The Sensitivity Report

To generate such a report, select “Sensitivity” in the output dialog box. The *sensitivity report* is generated in a separate tab and will resemble **Table 1**.

Table 1
Sensitivity Report Output

Adjustable Cells						
Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
\$C\$3	Barrels to process (in thousands) of light crude	300	0	10	5	1.666666667
\$D\$3	Barrels to process (in thousands) of heavy crude	600	0	12.5	2.5	4.166666667
Constraints						
Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$E\$10	Distillation used	1,800	0	3,000	1E+30	1,200
\$E\$11	Conversion used	2,400	3.125	2,400	480	800
\$E\$12	Blending used	2,400	1.25	2,400	1,200	400

There are two sections in the report: adjustable cells and constraints. The adjustable cells section gauges the sensitivity of the decision variables while the constraints section gauges the sensitivity of the constraints. We will focus on the latter section to answer Question 3.

Constraints

The report indicates the following information about the constraints (excluding non-negativity):

Final value—This represents the total amount used of the constraint resource.

Shadow price—This represents the monetary value gained from expanding a resource’s capacity. It tells us by how much we can improve the objective function value (contribution, in this case), by adding one unit of capacity to the resource. The shadow price will be greater than zero only if the constraint is being fully used. In order for a firm to justify adding capacity to a resource, the shadow price must be higher than the actual price of expansion. For instance, consider the distillation constraint. Distillation needs only 1,800 of its available 3,000 hours. Any additional availability will not improve contribution. The value of an extra unit of distillation is zero, which is indicated with a zero shadow price.

On the other hand, conversion uses all of its available 2,400 hours. Therefore, additional conversion hours could generate additional contribution. The shadow price of 3.125 tells us that if the conversion equipment availability increased from 2,400 to 2,401, we earn $\$3.125 \times 1,000 = \$3,125$ in additional contribution per week. (Note that we multiply by 1,000 because the total contribution has been expressed

in U.S. thousands.) Likewise, if the availability of conversion fell from 2,400 to 2,399, the total contribution would be reduced by \$3,125 from its current value of \$10,500,000. Keep in mind that these marginal values are calculated assuming that all other parameters are fixed at their current values.

Now that we know conversion has a shadow price, to determine the value of adding 100 conversion hours, we must confirm whether 100 hours falls with the constraint's *allowable range*. The *allowable increase* and *allowable decrease* figures in **Table 1** tell us the range in which the shadow price is valid. Within the allowable range, we can increase (decrease) the constraint's availability from its current value and still increase (decrease) the objective by the shadow price for each additional unit. If Petro added 100 hours to conversion, the 100 hours would fit within conversion's allowable increase of 480 hours. We can therefore improve contribution by \$3,125 per hour for up to 480 hours. The value to the firm of adding 100 hours of conversion is $\$3,125 \times 100 = \$312,500$. The market price of adding 100 conversion hours must be less than \$312,500 in order to justify the cost.

Consider the blending constraint. We can increase availability from 2,400 to $2,400 + 1,200 = 3,600$ hours per week and increase the total contribution by $1,200 \times 1,250 = \$1,500,000$. However, an increase from 3,600 to 3,601 will increase contribution by a different amount than the shadow price of 1,250. This increase will most likely be less than 1,250 (never greater than 1,250). Similarly, if blending were to decrease from 2,400 to $2,400 - 400 = 2,000$ hours, contribution would decrease by $400 \times 1,250 = \$500,000$. Any further decrease from 2,000 hours would reduce contribution by a different amount per additional unit, which would most likely be larger than 1,250 and never less than 1,250. Note that the allowable range is calculated assuming all other parameters stay fixed.

Adjustable Cells

To answer Question 4, we look at the adjustable cells section of **Table 1**, which lists each variable's:

- *Final value*: The optimal production amount (300 for light, 600 for heavy)
- *Objective coefficient*: Each product's current contribution (\$12.5 for heavy, \$10 for light)
- *Reduced cost*: The minimum amount the objective coefficient would need to change to cause the decision variable to become non-zero. In our case, the reduced cost is zero for both heavy and light because both products are being produced.
- *Allowable increase and decrease*: The range in which the solution stays the same. If the change to the object coefficient falls within each variable's allowable increase or decrease, then the optimal solution does not change, and the new objective function can be calculated. (If the value falls outside the range, then we must re-solve.)

Table 1 demonstrates that a \$1 decrease falls within heavy crude's allowable decrease of 4.16666667. Therefore, the optimal solution is still 300 for light and 600 for heavy, and the new objective function can be calculated as $(300,000 \times \$10) + (600,000 \times \$11.5) = \$9,900,000$.



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